

**UNIVERSITY OF THE FREE STATE**

**BLOEMFONTEIN CAMPUS**

**CSIS3744**

**MEMORANDUM**

**DEPARTMENT: COMPUTER SCIENCE AND  
INFORMATICS**

**CONTACT NUMBER: 051 401 3702**

**SICK TEST  
4<sup>th</sup> November 2021**

**ASSESSOR(S): 1. Prof Okuthe P. Kogeda**

**MODERATOR(S): 1. Mr W.S.J. Marais**

**TIME: 60 min**

**MARKS: 50**

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**INSTRUCTIONS:** Answer ALL the questions. Number your questions clearly and correctly.

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**Question 1 [17 Marks]**

Discuss the difference between the following terms and concepts:

- (a) Logical and Physical topology. [2 Marks]

<<< The term physical topology, or network topology, mostly applies to hardware and describes how computers, other devices, and cables fit together to form the physical network.

✓ The term logical topology has to do with software and describes how access to the network is controlled, including how users and programs initially gain access to the network and how specific resources, such as applications and databases, are shared on the network. ✓>>>

- (b) Unmanaged switches and managed switches. [2 Marks]

<<< An unmanaged switch provides plug-and-play simplicity with minimal configuration options and has no IP address assigned to it. Unmanaged switches are not very expensive, but

their capabilities are limited and they cannot support VLANs. ✓ Managed switches, on the other hand, can be configured via a command-line interface or a web-based management GUI, and sometimes can be configured in groups. Usually, they are also assigned IP addresses for the purpose of continued management. VLANs can only be implemented through managed switches, whose ports can be partitioned into groups. ✓>>>

(c) Hashing and Encryption. [2 Marks]

<<< Hashing means to transform data through an algorithm that generally reduces the amount of space needed for the data. Hashing is not the same thing as encryption, though it's often listed as a type of encryption and does, in a similar manner, transform data from one format to another. ✓ Encrypted data can be decrypted, but hashed data cannot. Hashing is mostly used to ensure data integrity-that is, to verify the data has not been altered, which is similar to the purpose of a checksum. ✓>>>

(d) Authentication and Authorization. [2 Marks]

<<<\* authentication-Authentication, in this case, user authentication, is the process of verifying a user's credentials (typically a username and password) to grant the user access to secured resources on a system or network. In other words, authentication asks the question, "Who are you?" ✓

\* authorization-Once a user has access to the network, the authorization process determines what the user can and cannot do with network resources. In other words, authorization asks the question, "What are you allowed to do?" Authorization restrictions affect Layer 2 segmentation, Layer 3 filtering, and Layer 7 entitlements. For example, what VLAN are you assigned to? What servers or databases can you access? What commands can you run on a device? ✓>>>

(e) A system fault and a system failure. [2 Marks]

<<< The difference between failures and faults is as follows:

\* failure-A deviation from a specified level of system performance for a given period of time. In other words, a failure occurs when something doesn't work as promised or as planned. For example, if your car breaks down on the highway, you can consider the breakdown to be a failure. ✓

\* fault-A malfunction of one component of a system. A fault can result in a failure. For example, the fault that caused your car to break down might be a leaking water pump. The goal of fault-tolerant systems is to prevent faults from progressing to failures. ✓>>>

(f) SNAT (Static Network Address Translation) and DNAT (Destination Network Address Translation). [2 Marks]

<<<\* SNAT-Using SNAT (Static Network Address Translation or Source Network Address Translation), the gateway assigns the same public IP address to a host each time it makes a request to access the Internet. Small home networks with only a single public IP address provided by its ISP use SNAT. ✓

\* DNAT or Destination NAT-Using DNAT (Destination Network Address Translation), hosts outside the network address a computer inside the network by a predefined public IP address. When a message sent to the public IP address reaches the router managing DNAT, the destination IP address is changed to the private IP address of the host inside the network. The router must maintain a translation table of public IP addresses mapped to various hosts inside the network. ✓>>>

(g) Single mode fiber (SMF) and Multimode fiber (MMF). [2 Marks]

<<< SMF (single mode fiber) consists of a narrow core of 8 to 10 microns in diameter. Laser-generated light travels a single path over the core, reflecting very little. Because it reflects little, the light does not disperse as the signal travels along the fiber. This continuity allows SMF to accommodate the highest bandwidths and longest distances (without requiring repeaters) of all network transmission media. ✓ MMF (multimode fiber) contains a core with a larger diameter than SMF, usually 50 or 62.5 microns, over which many pulses of light generated by a laser or LED light source travel at various angles. Signals traveling over multimode fiber experience greater attenuation than those traversing single mode fiber. Therefore, MMF is not suited to distances longer than a few kilometers. On the other hand, MMF is less expensive to install and, therefore, typically used to connect routers, switches, and servers on the backbone of a network or to connect a desktop workstation to the network. ✓>>>

(h) Unicast, Multicast and Anycast addresses. [3 Marks]

<<< A unicast address specifies a single node on a network. ✓ This differs from a multicast address, which is delivered to all nodes in a targeted, multicast group. ✓ An anycast address can identify multiple destinations, with packets delivered to the closest destination. For example, a DNS name server might send a DNS request to a group of DNS servers that have all been assigned the same anycast address. A router handling the request examines routes to all the DNS servers in the group and routes the request to the closest server. ✓>>>

## Question 2 [18 Marks]

Table I shows the OSI 7 layer reference model. The layer numbers are provided from 1 to 7. However, the corresponding OSI layers column, the hardware column and protocols column are not populated in the table. The OSI 7 layer reference model describes an idealized network communications protocol family. TCP/IP does not correspond to this model directly, as it either combines several OSI layers into a single layer, or does not use certain layers at all.

Table I: TCP/IP protocol stack

| Layer number | OSI Layers                  | Hardware                                                     | Protocols                                                                                             |
|--------------|-----------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 7            | Application <sup>1/2</sup>  |                                                              | HTTP, <sup>1/2</sup> FTP, <sup>1/2</sup> SMTP, <sup>1/2</sup> RIP, <sup>1/2</sup> SNMP <sup>1/2</sup> |
| 6            | Presentation <sup>1/2</sup> |                                                              |                                                                                                       |
| 5            | Session <sup>1/2</sup>      |                                                              |                                                                                                       |
| 4            | Transport <sup>1/2</sup>    | Firewall <sup>1/2</sup>                                      | UDP <sup>1/2</sup>                                                                                    |
| 3            | Network <sup>1/2</sup>      | Router <sup>1/2</sup>                                        | IP, <sup>1/2</sup> ARP, <sup>1/2</sup> ICMP <sup>1/2</sup>                                            |
| 2            | Data link <sup>1/2</sup>    | Bridge, <sup>1/2</sup> Access point <sup>1/2</sup>           | PPP <sup>1/2</sup>                                                                                    |
| 1            | physical <sup>1/2</sup>     | Hub, <sup>1/2</sup> NIC, <sup>1/2</sup> Cable <sup>1/2</sup> |                                                                                                       |

(a) Populate the table with the corresponding OSI layers. [3.5 Marks]

<<< see completed table section >>>

(b) Place the *firewall*, *router*, *bridge*, *access point*, *NIC*, *hub* and *Cable* hardware in their respective layers. [3.5 Marks]

<<< see completed table section >>>

(c) Place the *ARP*, *RIP*, *SNMP*, *PPP*, *UDP*, *IP*, *ICMP*, *HTTP*, *FTP*, *SMTP* protocols in their respective layers. [5 Marks]

<<< see completed table section >>>

(d) What is the responsibility of the data link layer of the OSI model? [2 Marks]

<<<The data link layer is responsible for multiplexing data streams, ✓ data frame detection, ✓ medium access, ✓ and error control. ✓ It ensures reliable point-to-point and point-to-multipoint connections in a communication network. ✓ Any two >>>

(e) As a network specialist, you decided to implement the bus topology using UTP cable for cabling.

(i) Explain the reasons for choosing to implement the bus topology. [2 Marks]

<<< Easy to connect a computer or peripheral to a linear bus; ✓ Requires less cable length than a star topology; ✓ >>>

(ii) Explain reasons for choosing UTP cable. [2 Marks]

<<< *less expensive, ✓ Easier to maintain and install. ✓* >>>

### **Question 3 [15 Marks]**

- (a) Explain some of the different types of wireless topologies that can be created?

[3 Marks]

<<< Some of the different types of wireless topologies are as follows:

\* ad hoc-A small number of nodes closely positioned transmit directly to each other without an intervening connectivity device. ✓

\* infrastructure-An intervening connectivity device, a WAP (wireless access point) or AP (access point), accepts wireless signals from multiple nodes and retransmits them to the rest of the network. To cover its intended range, an access point must have sufficient power and be strategically placed so that all connected nodes can communicate with it. ✓

\* mesh-Several access points work as peer devices on the same network, where the AP devices cooperate to provide more fault tolerant network access to clients. A wireless controller might be used only initially to configure the APs, or the APs might remain connected to the wireless controller for continued management. ✓>>>

- (b) Describe the TLS/SSL handshake process as initiated by a web client accessing a secure website. [3 Marks]

<<< Given the scenario of a browser accessing a secure Web site, the SSL/TLS handshake works as follows:

1. The browser, representing the client computer in this scenario, sends a client\_hello message to the Web server, which contains information about what level of security the browser is capable of accepting and what type of encryption the browser can decipher. The client\_hello message also establishes a randomly generated number that uniquely identifies the client and another number that identifies the SSL session. ✓

2. The server responds with a server\_hello message that confirms the information it received from the browser and agrees to certain terms of encryption based on the options supplied by the browser. Depending on the Web server's preferred encryption method, the server may choose to issue to the browser a public key or a digital certificate. ✓

3. If the server requests a certificate from the browser, the browser sends it. Any data the browser sends to the server is encrypted using the server's public key. Session keys used only for this one session are also established. ✓>>>

- (c) Explain any FOUR reasons for using VLANs. [4 Marks]

<<< Reasons for using VLANs include:

\* You wish to isolate connections with heavy or unpredictable traffic patterns, such as when separating heavy VoIP traffic from other network activities. ✓

\* You want to identify groups of devices whose data should be given priority handling, such as executive client devices or an ICS (industrial control system) that manages a refrigeration system or a gas pipeline. ✓

\* You want to contain groups of devices that rely on legacy protocols incompatible with the majority of the network's traffic, such as a legacy SCADA (supervisory control and data acquisition) system monitoring an oil refinery. ✓

\* You wish to separate groups of users who need special or limited security or network functions, such as when setting up a guest network. ✓

\* You are configuring temporary networks, such as when making specific network resources available to a short-term project team. ✓

\* You intend to reduce the cost of networking equipment, such as when upgrading a network design to include additional departments or new types of network traffic. ✓>>>

- (d) Discuss three of the more common network performance metrics that are utilized? [3 Marks]

<<< *Some of the more common performance metrics are as follows:*

- 1) Utilization-This metric refers to the actual throughput used as a percentage of available bandwidth. No network should operate at maximum capacity. Identify patterns of utilization and ensure that available bandwidth accounts for utilization spikes.✓
- 2) Error rate-Bits can be damaged in transit due to EMI or other interference. The calculated percentage of how often this occurs is the error rate.✓
- 3) Packet drops-Packets that are damaged beyond use, arrive after their expiration, or are not allowed through an interface are dropped. Packet drops result in delayed network communications while devices wait for responses or have to resend transmissions. Knowing what's normal for your network will help you identify problems when packet drop rates vary.✓
- 4) Jitter-All packets experience some latency. When successive packets experience varying amounts of latency, resulting in their arriving out of order, the user experience is degraded. This is called jitter, a problem that can be addressed through traffic management techniques.✓>>>
- (e) Calculate the subnet mask that can be used to segment the 172.16.0.0 network to allow for a minimum of 6 subnets while maximizing the number of hosts per subnet? [2 Mark]
- <<< 255.255.224.0 ✓>>>

**End of Paper**